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**B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination:
October/November - 2025**

Advanced Python Programming (G06-DSC2-0304)

Day & Date: Monday, 03-11-2025
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose the Correct Alternative.

06

- 1) In Python, what does *args allow in a function definition?
 - a) Keyword-only arguments.
 - b) A variable number of positional arguments.
 - c) Default argument values.
 - d) Only keyword arguments
- 2) Which re module function returns all non-overlapping matches of a pattern in a string as a list?

a) search()	b) split()
c) sub()	d) findall()
- 3) What is the difference between shallow copy and deep copy in Python?
 - a) Shallow copy copies nested objects recursively; deep copy doesn't.
 - b) Shallow copy doesn't copy nested objects; deep copy does recursively.
 - c) Shallow copy is faster than deep copy always.
 - d) Deep copy is used for immutable objects only
- 4) How do you convert a Python dictionary to a JSON string?

a) json.load(dictionary)	b) json.dump(dictionary)
c) json.dumps(dictionary)	d) json.read(dictionary)
- 5) Which Python library is commonly used for reading and writing Excel files (.xlsx)?

a) openpyxl	b) csv
c) xlrd	d) pandas
- 6) What is a closure in Python?
 - a) A function defined inside another function retaining outer scope variables.
 - b) A global variable is accessible everywhere.
 - c) A built-in Python data structure.
 - d) A class method.

- Q.2 Answer the following questions. (Any three) 06**
- a) Differentiate between global and nonlocal keywords in Python functions.
 - b) What is the purpose of the pillow module when working with Excel files in Python?
 - c) What is the benefit of using the with statement for file handling in Python?
 - d) What is garbage collection in python?
 - e) What is the difference between a workbook and a worksheet in Excel terminology?
- Q.3 Answer the following questions. (Any two) 06**
- a) Explain the concept of closures in Python with an example.
 - b) Explain the use of \d, \w, and \s sequence characters in Reg. Ex with illustrative patterns.
 - c) How can you manage (add, remove, rename) sheets in an Excel workbook using openpyxl?
- Q.4 Answer the following questions. (Any two) 06**
- a) What are the steps to create a bar chart in an Excel spreadsheet using openpyxl?
 - b) Explain the use of grouping () in regular expressions with an example.
 - c) Explain type hinting in Python with an example. What are its benefits?
- Q.5 Answer the following question. (Any one) 06**
- a) Explain the process of serialization and deserialization of data using the json module in Python. Provide examples of converting Python dictionaries to JSON strings and files, and vice versa.
 - b) Describe the concepts of iterables, iterators, and generators in Python. Provide examples illustrating their differences